

Simple Data Pipeline Documentation

This project demonstrates a simple modern data pipeline using SQL transformations and Power BI reporting.

1. Project Overview

The project uses transaction data to create a structured data pipeline. Raw data is cleaned, transformed into customer metrics, and finally converted into a business-ready monthly revenue report.

2. Data Flow Diagram

transactions table → stg_orders → int_customer_metrics → mart_monthly_revenue → Power BI Dashboard

3. Model Descriptions

Model 1: stg_orders.sql (Staging)

- Cleans raw transaction data
- Converts order_date into standard DATE format
- Removes records with null order IDs

Model 2: int_customer_metrics.sql (Intermediate)

- Calculates total orders for each customer
- Calculates total spend per customer
- Finds the latest order date

Model 3: mart_monthly_revenue.sql (Mart)

- Creates monthly revenue summary
- Compares current month revenue with previous month
- Calculates revenue difference

4. Data Quality Tests

- Unique test on customer_id
- Not-null test on order_date
- Revenue validation where revenue > 0

5. Dashboard Information

The Power BI dashboard displays:

- Total Orders
- Total Spend
- Revenue Change
- Monthly Revenue Trend
- Last Month vs Current Month Revenue Comparison

6. Execution Steps

1. Create and use the sales_data database
2. Run staging model query
3. Run intermediate model query
4. Run mart model query
5. Execute validation tests
6. Connect mart table to Power BI dashboard

7. Conclusion

This project demonstrates a simple ETL workflow using SQL for data transformation and Power BI for visualization.